

Linear Algebra for Data Science

Tutorial 2

Orthogonal bases, orthogonal matrices, and QR

This sheet covers both sessions of Module 2. Sections 1–3 follow Session 2.1 (orthogonal matrices, Gram–Schmidt, QR , least squares); Section 4 is where QR earns its keep; Section 5 collects the three parts of Session 2.2 that Lab 1 will not reach. Starred items (★) are optional.

1. From the Normal Equation to $R\mathbf{x} = Q^\top \mathbf{b}$

Questions

- Q1. In the reduced factorization $A = QR$ with A of size $m \times n$, $m > n$, the matrix Q has orthonormal *columns* but is not square. Which of $Q^\top Q = I$ and $QQ^\top = I$ holds? What is the other product, geometrically?
- Q2. The derivation of $R\hat{\mathbf{x}} = Q^\top \mathbf{b}$ cancels $R^\top$ from $R^\top R\hat{\mathbf{x}} = R^\top Q^\top \mathbf{b}$. Exactly which hypothesis licenses the cancellation, and where would the argument break for a rank-deficient A ?
- Q3. Both routes to $\hat{\mathbf{x}}$ — the normal equation and $R\mathbf{x} = Q^\top \mathbf{b}$ — produce the same vector in exact arithmetic. State, in one sentence each, (a) why they do, and (b) why they are nonetheless not interchangeable.
- Q4. The lecture wrote $A^\dagger = R^{-1}Q^\top$ and immediately added that you still never compute it. What *is* computed instead, and what would forming R^{-1} explicitly cost you — in flops, and in accuracy?

Tasks

- T1. Show that solving $R\mathbf{x} = Q^\top \mathbf{b}$ for the reduced factorization $A = QR$ is equivalent to the normal equation in exact arithmetic, then say why the two are nonetheless not interchangeable in floating point.
- T2. Fit a straight line $y = c + dt$ to the data $t = 1, 2, 3, 4$, $\mathbf{b} = (1, 3, 4, 6)^\top$, so that

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} c \\ d \end{pmatrix}.$$

- (a) Run Gram–Schmidt on the two columns *without normalising*, obtaining $W = (\mathbf{w}_1 \ \mathbf{w}_2)$ and $D = \text{diag}\{\|\mathbf{w}_1\|, \|\mathbf{w}_2\|\}$. Then get Q and R by the lecture’s lifhack: divide the k^{th} column of W , and the k^{th} row of $W^\top A$, by $\|\mathbf{w}_k\|$.
- (b) Compute $Q^\top \mathbf{b}$ and solve $R\mathbf{x} = Q^\top \mathbf{b}$ by back substitution.
- (c) Solve the normal equation $A^\top A\mathbf{x} = A^\top \mathbf{b}$ directly and confirm you get the same $\hat{\mathbf{x}}$.
- (d) Compute the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ and verify $\mathbf{r} \perp \mathbf{w}_1$, $\mathbf{r} \perp \mathbf{w}_2$. What is $\|\mathbf{r}\|^2$?
- T3. Continue with A from T2. The full factorization $A = \tilde{Q}\tilde{R}$ has $\tilde{Q}$ of size 4×4 .
- (a) The lecture claims the first two columns of $\tilde{Q}$ span $\text{col}(A)$ and the remaining two span $\mathcal{N}(A^\top)$. Prove the second claim in general: if the columns of $\tilde{Q}$ are an ONB of $\mathbb{R}^m$ whose first n span $\text{col}(A)$, why must the rest span $\mathcal{N}(A^\top)$?
- (b) Find an orthonormal basis of $\mathcal{N}(A^\top)$ for this A by solving $A^\top \mathbf{y} = \mathbf{0}$ directly, and check your two vectors are orthogonal to $\mathbf{q}_1$ and $\mathbf{q}_2$.
- (c) Verify Pythagoras on $\mathbf{b}$: decompose $\mathbf{b} = P\mathbf{b} + (I - P)\mathbf{b}$ with $P = QQ^\top$, and check $\|\mathbf{b}\|^2 = \|P\mathbf{b}\|^2 + \|\mathbf{r}\|^2$.

2. Gram–Schmidt: One Factorization, Two Arithmetics

Questions

- Q1. Classical and modified Gram–Schmidt are *identical* in exact arithmetic. Prove it for $k = 3$: show that subtracting the projection of the running residual $\mathbf{w}_3^{(1)}$ onto $\mathbf{w}_2$ gives the same vector as subtracting the projection of the original $\mathbf{v}_3$ onto $\mathbf{w}_2$.
- Q2. The QR factorization of a full-column-rank A is unique up to the signs of the columns of Q . Since Gram–Schmidt and Householder therefore compute *the same* object, in what sense can one of them be “better”? Answer in one sentence, and name the quantity that measures the difference.
- Q3. When $\mathbf{v}_k$ nearly lies in the span of $\mathbf{w}_1, \dots, \mathbf{w}_{k-1}$, the residual $\mathbf{w}_k$ is small compared with the terms that produce it. Explain why this corrupts the *direction* of $\mathbf{w}_k$ rather than its length, and why direction is the only thing we wanted.
- Q4. Suppose Q has lost orthogonality, so $Q^\top Q = I + E$ with $\|E\| = 0.5$. What goes wrong with $P := QQ^\top$? Is it still symmetric? Still idempotent? Still a projector onto anything?

Tasks

- T1. (The 60°, by hand.) Let $\varepsilon = 10^{-3}$ and

$$A = \begin{pmatrix} 1 & 1 & 1 \\ \varepsilon & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}.$$

Work throughout in decimal arithmetic carrying **six significant digits**, rounding after every operation. [So $1 + \varepsilon^2 = 1.000001$ is stored as 1.00000. This is a scale model of double precision, where the same thing happens at $\varepsilon = 10^{-9}$.]

- (a) *Predict first, in writing:* the three columns are very nearly parallel. Do you expect the computed $\mathbf{q}_2, \mathbf{q}_3$ to come out orthogonal? If not, how badly?
- (b) Run *classical* Gram–Schmidt by hand and compute $\mathbf{q}_2 \cdot \mathbf{q}_3$.
- (c) Run *modified* Gram–Schmidt by hand on the same matrix and compute $\mathbf{q}_2 \cdot \mathbf{q}_3$ again.
- (d) Locate the single rounding that causes the difference. Which subtraction cancels, and what information is destroyed?
- T2. Take A from T1 with $\varepsilon = 10^{-8}$ and check the same three quantities at a computer: build Q by classical Gram–Schmidt, by modified Gram–Schmidt, and by `numpy.linalg.qr`, and print $\|Q^\top Q - I\|$ for each. Print the full matrix $Q^\top Q$, not only its distance from I — the off-diagonal entries are the story. Compare with `cond(A)` from `numpy.linalg.cond`.
- T3. * Show that classical Gram–Schmidt applied *twice* (orthogonalise, then orthogonalise the result again against the same $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$) recovers orthogonality to working precision. Explain the folklore “twice is enough, and thrice is pointless” in terms of the mechanism you identified in T1(d). *

3. Building Q from Reflections and Rotations

Questions

- Q1. For $\mathbf{u} \neq \mathbf{0}$ let $Q_{\mathbf{u}} = I - 2\mathbf{u}\mathbf{u}^\top / \mathbf{u}^\top \mathbf{u}$. Show that $Q_{\mathbf{u}}$ is symmetric, orthogonal, and satisfies $Q_{\mathbf{u}}^2 = I$. What is $Q_{\mathbf{u}}\mathbf{u}$, and what is $Q_{\mathbf{u}}\mathbf{v}$ for $\mathbf{v} \perp \mathbf{u}$?

- Q2. Householder reflectors have $\det = -1$ and Givens rotations have $\det = +1$. Deduce that a product of k reflectors is a rotation exactly when k is even, and say what this means for the $\tilde{Q}$ produced by a Householder sweep of an $m \times n$ matrix.
- Q3. In practice one never forms $Q_{\mathbf{u}}$ as a matrix. Given $\mathbf{u}$ and a vector $\mathbf{v}$, write down how to compute $Q_{\mathbf{u}}\mathbf{v}$, and compare the flop count with the m^2 of a matrix–vector product.
- Q4. Givens needs more rotations than Householder needs reflectors, yet is preferred on sparse and on nearly-triangular matrices. Explain the apparent contradiction, and name one streaming application where Givens is the only sensible choice.

Tasks

- T1. **(Both signs, and why one of them fails.)**
- Let $\mathbf{x} = (3, 4)^\top$. Compute both admissible reflection vectors $\mathbf{u}_{\pm} = \mathbf{x} \mp \|\mathbf{x}\|\mathbf{e}_1$, form $Q_{\mathbf{u}_+}\mathbf{x}$ and $Q_{\mathbf{u}_-}\mathbf{x}$ by hand, and confirm each is $\pm 5\mathbf{e}_1$.
 - Now let $\mathbf{x} = (1, \delta)^\top$ with $\delta = 10^{-9}$, in double precision. Show that the “natural” choice $\mathbf{u} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1$ computes as $\mathbf{u} = (0, \delta)^\top$, and that the resulting $Q_{\mathbf{u}}\mathbf{x}$ is **not** collinear with $\mathbf{e}_1$. What is it?
 - Repeat with $\mathbf{u} = \mathbf{x} + \text{sign}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ and verify that $Q_{\mathbf{u}}\mathbf{x} = -\|\mathbf{x}\|\mathbf{e}_1$ exactly, with no cancellation anywhere.
- T2. **(Updating a factorization with one new observation.)** Return to the line fit of Section 1 T2, where $R = \begin{pmatrix} 2 & 5 \\ 0 & \sqrt{5} \end{pmatrix}$ and $Q^\top \mathbf{b} = (7, 8/\sqrt{5})^\top$. A fifth data point arrives: $t_5 = 5$, $b_5 = 8$, i.e. a new row $\mathbf{a}^\top = (1, 5)$.
- Form the 3×2 matrix $\begin{pmatrix} R \\ \mathbf{a}^\top \end{pmatrix}$ and use two Givens rotations to return it to upper-triangular form R' . [G_1 acts on rows 1 and 3 to zero the $(3, 1)$ entry; G_2 acts on rows 2 and 3.] Verify $R'^\top R' = A'^\top A'$ for the enlarged design A' .
 - Apply the same two rotations to $(7, 8/\sqrt{5}, b_5)^\top$ and solve the updated triangular system. Check against the normal equation for all five points.
 - The third component of the rotated right-hand side is not used in the back substitution. Square it. What does it equal, and what is the total residual $\|\mathbf{r}'\|^2$ for the five-point fit?
- T3. * Count. For A of size $m \times n$ with $m \gg n$: how many Householder reflectors does a full sweep use, and how many Givens rotations would the same job need? Then suppose A is upper Hessenberg (one non-zero subdiagonal). Redo both counts and state which method you would call. *

4. What QR Is For

Questions

- Q1. With column pivoting, QR factorizes $AP = QR$ for a permutation P chosen so that $|r_{11}| \geq |r_{22}| \geq \dots$. Explain why a sharp drop in $|r_{kk}|$ signals that the columns are nearly dependent, and why this is a cheap substitute for inspecting the singular values.
- Q2. Why is $|\det A| = \prod_k |r_{kk}|$, and why is this a better way to compute a determinant than cofactor expansion — both in cost and in accuracy?
- Q3. In the orthogonal Procrustes problem $\min_Q \|XQ - Y\|_F$ over orthogonal Q , what would happen to the minimum if Q were allowed to be any invertible matrix? What does the orthogonality constraint encode about the modelling assumption?

- Q4. Two word embeddings are trained on two corpora. Name what must be true of the two data matrices before Procrustes alignment is even a well-posed question. [Look carefully at the lecture's phrase “row j of X and row j of Y are the same point”.]

Tasks

- T1. **(Effective rank, at a computer.)** Build $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \\ 1 & 5 & 6 \end{pmatrix}$ and then perturb it: $A_\eta := A + \eta N$ with N a fixed random matrix of unit norm and $\eta = 10^{-10}$.
- What is $\text{rank } A$ exactly? Say why before computing.
 - Compute the pivoted factorization with `scipy.linalg.qr(A_eta, pivoting=True)` and print `diag R`. Where is the drop?
 - Compare with the singular values from `numpy.linalg.svd`. Do the two diagnostics agree on where to cut?
 - Set $\eta = 10^{-2}$ and repeat. Is the rank decision still unambiguous? What does that tell you about the phrase “the rank of a data matrix”?
- T2. *** (Procrustes by hand, in 2-D.)** Let X have rows $(1, 0)$, $(0, 1)$, $(-1, 0)$ and let Y have rows $(0, 1)$, $(-1, 0)$, $(0, -1)$. Restrict Q to rotations $R_\alpha = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$.
- Write $\|XR_\alpha - Y\|_F^2$ as a function of α alone, using $\|M\|_F^2 = \text{trace}(M^\top M)$.
 - Minimise it by hand and identify the aligning rotation.
 - Show in general that $\|XQ - Y\|_F^2 = \|X\|_F^2 + \|Y\|_F^2 - 2\text{trace}(Q^\top X^\top Y)$ for orthogonal Q , so that the minimisation is a *maximisation* of $\text{trace}(Q^\top X^\top Y)$. [This is the step the SVD finishes, in Module 5.]
- T3. In one paragraph each, say what QR buys you in three settings named but not developed in the lecture: (i) solving a square system $A\mathbf{x} = \mathbf{b}$; (ii) computing $|\det A|$; (iii) the eigenvalue iteration $A_k = Q_k R_k \mapsto A_{k+1} = R_k Q_k$. For (i), say why LU nonetheless remains the default.

5. Three Things Lab 1 Will Not Test

Session 2.2's Fourier material is met in code, in Lab 1. These three are not, so they are settled here.

Questions

- Q1. Let A be symmetric positive definite and $\langle \mathbf{x}, \mathbf{y} \rangle_A := \langle A\mathbf{x}, \mathbf{y} \rangle$. Verify the three inner-product axioms, and say exactly which axiom fails if A is merely positive semi-definite.
- Q2. The conjugate-gradient method expands the *unknown* solution $\hat{\mathbf{x}}$ in a conjugate basis, with coefficients $\alpha_k = \langle \hat{\mathbf{x}}, \mathbf{p}_k \rangle_A / \langle \mathbf{p}_k, \mathbf{p}_k \rangle_A$. Explain in two lines why this is not circular.
- Q3. CG never needs A as a stored matrix. Which single operation does it need instead, and why does that make it the method of choice for a graph Laplacian with 10^7 vertices?
- Q4. In the PCA reduction, why is $\hat{\mathbf{x}}_j = P_L \mathbf{x}_j$ *forced* rather than chosen? Name the result from Session 1.1 that forces it.

Tasks

- T1. **(Conjugate is not perpendicular.)** Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{b} = (3, 0)^\top$.
- Starting from $\mathbf{e}_1, \mathbf{e}_2$, run Gram–Schmidt in $\langle \cdot, \cdot \rangle_A$ to produce a conjugate pair $\mathbf{p}_1, \mathbf{p}_2$. Verify $\langle \mathbf{p}_1, \mathbf{p}_2 \rangle_A = 0$.
 - Compute the ordinary Euclidean angle between $\mathbf{p}_1$ and $\mathbf{p}_2$. It is not 90° . Sketch both vectors together with an A -unit ball $\{\mathbf{x} : \langle \mathbf{x}, \mathbf{x} \rangle_A = 1\}$ and say what “perpendicular” means in that picture.
 - Compute α_1, α_2 from $\alpha_k = \langle \mathbf{b}, \mathbf{p}_k \rangle / \langle \mathbf{p}_k, \mathbf{p}_k \rangle_A$ and assemble $\hat{\mathbf{x}} = \alpha_1 \mathbf{p}_1 + \alpha_2 \mathbf{p}_2$. Check $A\hat{\mathbf{x}} = \mathbf{b}$.
 - You solved a 2×2 system in exactly 2 steps, with no inverse and no elimination — only scalar products. State what the analogous claim is for $n \times n$, and why in practice one stops long before step n .
- T2. **(PCA as a maximisation, with no spectral theorem.)** Four centred data points in $\mathbb{R}^2$: $(2, 1), (-2, -1), (1, -1), (-1, 1)$.
- Verify the mean is $\mathbf{0}$ and compute $S = \frac{1}{N} \sum_j \mathbf{x}_j \mathbf{x}_j^\top$.
 - For $k = 1$ the problem is $\max_{\|\mathbf{q}\|=1} \mathbf{q}^\top S \mathbf{q}$. Parametrise $\mathbf{q} = (\cos \theta, \sin \theta)^\top$, reduce $\mathbf{q}^\top S \mathbf{q}$ to the form $p + a \cos 2\theta + b \sin 2\theta$, and maximise by hand.
 - Report the optimal direction and the maximal value. Confirm that the value is the larger root of $\lambda^2 - \text{trace}(S)\lambda + \det(S) = 0$.
 - Explain, in one sentence, why $\min_L \sum_j \|\mathbf{x}_j - \hat{\mathbf{x}}_j\|^2$ and $\max_L \sum_j \|\hat{\mathbf{x}}_j\|^2$ are the same problem, naming the identity used.
- T3. *** (The DFT is unitary.)** Let $\omega = e^{2\pi i/N}$ and let F have entries $F_{jk} = \frac{1}{\sqrt{N}} \omega^{jk}$, $j, k = 0, \dots, N-1$. Show that the columns of F are orthonormal in the complex inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^* \mathbf{y}$, so F is unitary. [The pre-class lesson did $N = 4$ by hand; do the general case.] *